Name: ___________________________________

Unearthing Mars — A Historical Perspective

Student Directions

Focus Questions: While performing this investigation, use the following focus questions to guide your work.

1. How did Kepler's Laws affect the ability of scientists to determine the locations and the motions of objects in the solar system?
2. Which mathematical and/or computational representations are used to plot the location of Mars and predict its orbit?
3. How do gravitational effects change the orbits of objects in the solar system?
4. How can the principles of relative motion and perspective from Earth and other solar system objects be used to describe celestial phenomena today?

Purpose of Investigation: In the late 1500s, Tycho Brahe made several very accurate observations of the positions of Mars 20 years before the telescope was even invented. In fact, his measurements can be used to locate astronomical objects to this day. Johannes Kepler, Brahe's assistant, later determined the orbital period of Mars to be 687 Earth days. With Brahe's data and the orbital period of Mars, Kepler devised a method of triangulation, using his perspective from Earth to determine the location of Mars in its orbit.

Mars is a major focus for space exploration in the present day. The goal of the United States' exploration of Mars is to provide researchers with a continuous flow of discovery and scientific information, obtained from robotic orbiters, landers, and mobile laboratories interconnected by a sophisticated Mars/Earth communications network. The present-day planning and implementation of space probe missions to Mars utilize the same principles of relative motion and gravity that were investigated by Brahe in the late 1500s and by Kepler in the early 1600s.

Your tasks in **Session 1: Part 1** and **Session 1: Part 2 (*Student Answer Packet 1*)** will be to use mathematical and computational representations to:

- create a model of the orbit of Mars from given and derived calculations
- create a second model of the orbit of Mars using the same triangulation method as Johannes Kepler
- conduct a percent error analysis between the known eccentricity of Mars and the calculated eccentricity of Mars from the second predicted model of Mars' orbit in this investigation

Your task in **Session 2 (*Student Answer Packet 2*)** will be to apply what you learned from **Session 1** to identify patterns of evidence from the early history of astronomical observations of Mars and Earth and to use principles of relative motion, perspective, and gravity to describe the motions of solar system objects.

Materials (per student):

- 1 circular protractor
- 1 metric ruler
- 1 loop of string
- 2 push pins
- 1 corkboard (or similar)
- 4 different sharpened colored pencils
- 1 pencil sharpener
- 1 calculator
- 1 pencil sharpener

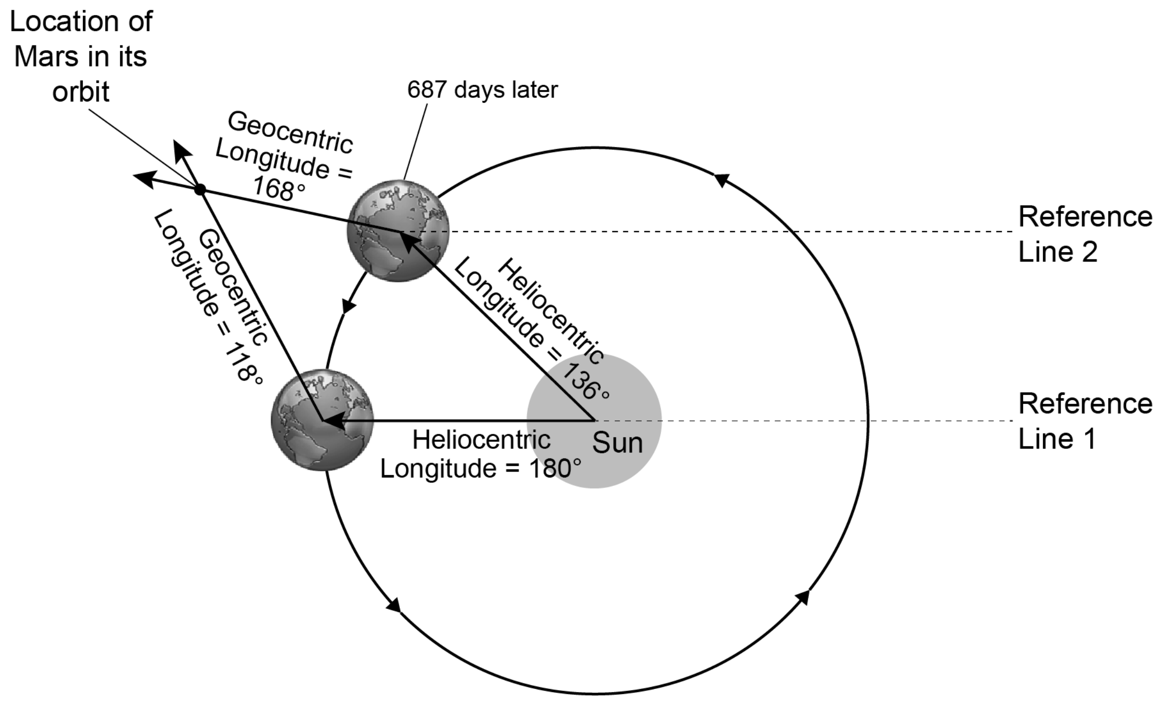
Summary: You will be using mathematical and computational representations to create a model of the orbit of Mars. You will then triangulate four locations to draw a predicted orbit of Mars using six plotted locations. Finally, you will calculate your percent error using the first orbit model and the second predicted orbit model of Mars.

Procedure:

Session 1: Part 1 and **Session 1: Part 2**

1. Write your name on ***Student Answer Packet 1***.
2. In ***Student Answer Packet 1***, the formula for eccentricity for Mars' orbit is missing one value, the distance between foci. The accepted value for Mars' eccentricity is <u>0.094</u>. Use algebra to solve for this missing value. Show all work in the space provided. Round your answer *to the nearest tenth*.
3. Remove and place the *Model of Mars' Orbit Using Given Calculations* worksheet (page 3 of ***Student Answer Packet 1***) on the board. On this worksheet, plot the second focus position to the left of the "Sun" (using the distance calculated in step 2) with a large dot and label this dot "Focus 2."
4. Place the pins in each of the dots representing the Sun and the second focus.
5. Place the string *under* the plastic of the pins to construct an ellipse (be sure to stretch the string tight with a pencil). Label this ellipse "Mars' Orbit."
6. Remove the pins from the paper.
7. Place one dot on your drawn "Mars' Orbit" at the point closest to the "Sun's" position. Label this dot "Perihelion."
8. Place one dot on your drawn "Mars' Orbit" at the point farthest from the "Sun's" position. Label this dot "Aphelion."

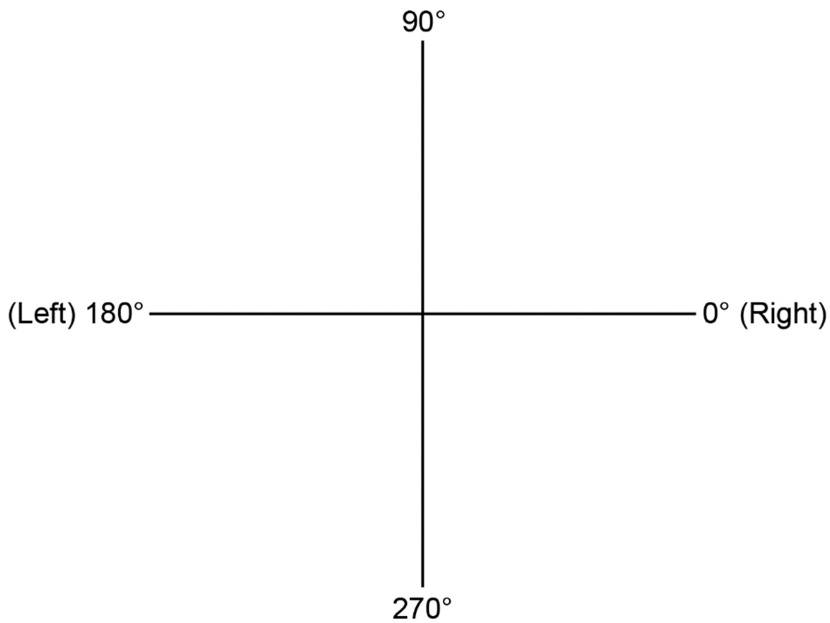
The model below illustrates the triangulation process for locating Mars by using two positions of Earth 687 Earth days apart. Two heliocentric longitudes show where Earth is relative to the Sun, and two geocentric longitudes are drawn and intersect to indicate the position of Mars outside of Earth's orbit. Reference lines for measuring the heliocentric longitude and geocentric longitude are included.



(Not drawn to scale)

9. Remove the board. Using the *Data Table: Triangulation of Mars* on page 6, plot the four Mars positions on the *Model of Mars' Orbit Using Triangulation* on page 4 of ***Student Answer Packet 1*** by following the steps below (each Mars Position will be triangulated using two dates 687 Earth days apart). The M1 and M2 Mars Positions (shaded in the data table) have already been plotted. You will plot Mars Positions M3 through M6.
   a. The data in the "Heliocentric Longitude of Earth" column will be plotted on Earth's orbit for Mars Position M3. Center the protractor on the Sun and align zero degrees with the Reference Line 1 (0°) Position. Plot the heliocentric longitude angle on Earth's orbit with a dot using a colored pencil.
   b. The data in the "Geocentric Longitude of Mars" column will be plotted outside Earth's orbit. Center the protractor on the plotted dot on Earth's orbit and align zero degrees parallel to the horizontal lines of the grid.

The protractor must be centered on the Sun dot and match the numbers on Earth's orbit.



Measure the geocentric longitude angle and draw a reference point <u>outside</u> Earth's orbit to indicate this angle. Then draw a line using the same colored pencil approximately seven centimeters in length from the point on Earth's orbit through the reference point <u>outside</u> Earth's orbit.

c. Repeat steps 9a and 9b for the second date for Mars Position M4. The drawn geocentric longitude lines will intersect at a position <u>outside</u> Earth's orbit. The intersection of the two geocentric longitude lines locates the planet Mars for that pair of observations. Indicate this intersection with a dot, using one of the colored pencils, and label it M4.

10a. Repeat steps 9a, 9b, and 9c for Mars Positions M5 and M6. Be sure to place labels on all four Mars Positions. <u>Each of the remaining Mars Position lines and dots should be a different color</u>.

10b. Compare your four Mars Positions with your partner.

11. Using your plotted points, draw the approximate orbit of Mars on your model so that the orbit passes through all six of the triangulated plotted Mars Positions. Label your orbit "Mars' Orbit."

12. Use the ruler to draw a line from Mars Position M1 to Mars Position M3 passing through the Sun. Label this line "Major Axis."

13. Measure the length of the Major Axis, in centimeters, *to the nearest tenth*. Record this value in ***Student Answer Packet 1***.

14. To find the distance between foci:
    a. Measure the length from M1 to the Sun, in centimeters, *to the nearest tenth*. Record this value in ***Student Answer Packet 1***.
    b. Measure the length from the Sun to M3, in centimeters, *to the nearest tenth*. Record this value in ***Student Answer Packet 1***.
    c. Using your measured values from 14a and 14b, determine the Aphelion and Perihelion positions of the orbit. Label these positions "Aphelion" and "Perihelion" on either end of the Major Axis on page 4 of ***Student Answer Packet 1***.
    d. Subtract the Perihelion value (Sun to M3 distance) from the Aphelion value (M1 to Sun distance) to determine the distance between foci for this model. Record this value, in centimeters, *to the nearest tenth* in ***Student Answer Packet 1***.
15. Calculate the eccentricity of Mars' orbit using the values from steps 13 and 14d. Show your work in the workspace and record this answer *to the nearest thousandth* in ***Student Answer Packet 1***.
16. Calculate the percent error using your eccentricity value from step 15 and the accepted eccentricity for Mars' orbit provided in step 2. Show your work in the workspace and record this answer *to the nearest tenth* in ***Student Answer Packet 1***.
17. Place all your materials back into the jumbo-sized plastic bags.
18. Return all your materials and paperwork to your teacher.

Session 2

Your task on **Session 2** is to apply what you learned in **Session 1** to independently answer the investigation questions found in ***Student Answer Packet 2***.

Data Table: Triangulation of Mars

Data	Mars Position	Date	Heliocentric Longitude of Earth (position of Earth relative to the Sun)	Geocentric Longitude of Mars (position of Mars relative to Earth)
Historical	M1	February 17, 1585	159°	135°
		March 15, 1587	115°	182°
	M2	March 28, 1587	196°	168°
		February 12, 1589	153°	218°
	M3	September 19, 1591	5°	284°
		August 6, 1593	323°	347°
	M4	December 7, 1593	86°	3°
		October 25, 1595	41°	49°
Modern	M5	May 26, 1935	245°	187°
		April 12, 1937	202°	246°
	M6	January 21, 1944	121°	66°
		December 8, 1945	76°	123°